

## 1. Executive Summary

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This document details the exploratory data analysis (EDA) and initial preprocessing tasks performed on Day 1 of the Insurance Premium Prediction project. The overarching goal of this project is to develop a robust machine learning pipeline capable of accurately predicting the 'Total\_Annual\_Premium\_INR' for properties based on various structural, locational, and demographic features. Day 1 focused heavily on data ingestion, structural inspection, handling of missing values, and a preliminary univariate analysis of the target variable.

## 2. Day 1 Objectives

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The key milestones planned and executed during this phase included:

- Environment setup and ingestion of the raw dataset using Pandas.
- High-level structural analysis to understand dimensions, datatypes, and feature categories.
- Descriptive statistical analysis of numerical variables to identify central tendencies and dispersion.
- Identification of data quality issues, specifically missing values and duplicate records.
- Univariate analysis of the target variable to understand its probability distribution.

## 3. Detailed Task Breakdown & Findings

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### 3.1 Data Ingestion and Structural Overview

The project commenced by importing necessary Python libraries (Pandas, NumPy, Matplotlib) and loading the 'dataset.csv' file. The dataset consists of exactly 10,000 observations (rows) and 22 distinct attributes (columns). A preview of the head of the dataframe confirmed the successful loading of diverse features such as Property\_Type, Location\_Type, and Property\_Age\_Years.

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Day 1 Comprehensive Report | July 28, 2026

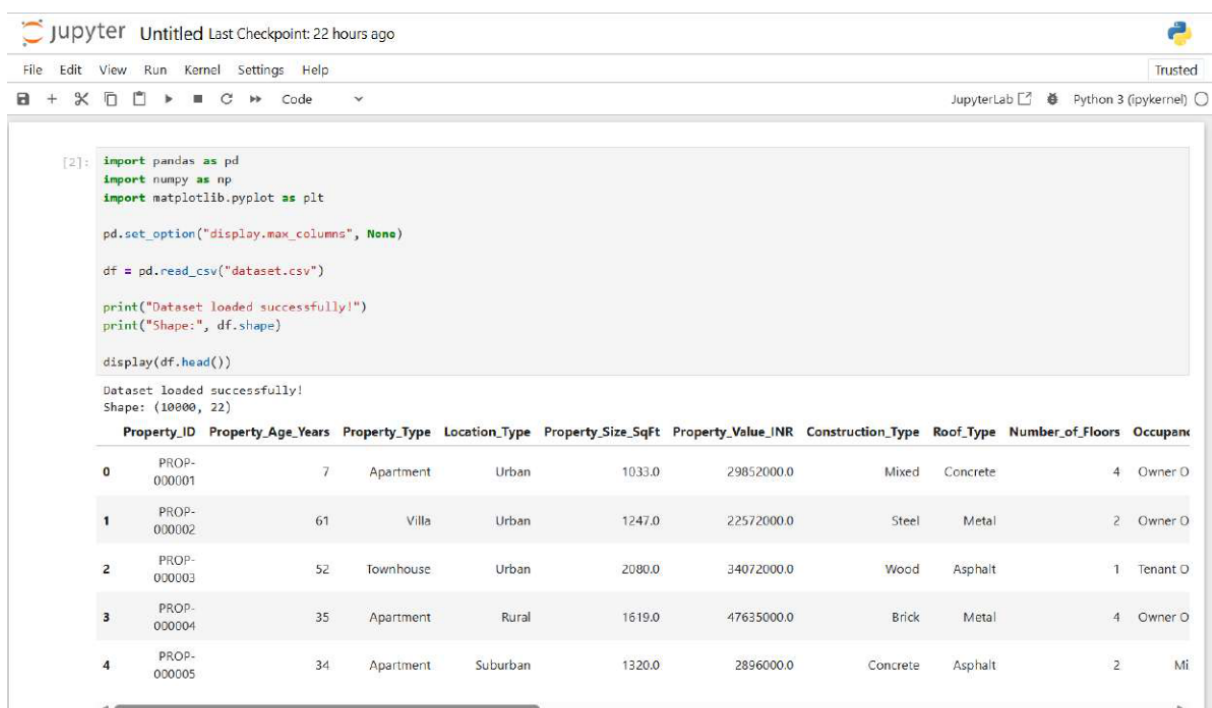


Fig 1: Library imports and initial data loading demonstrating the 10,000 x 22 structure.

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Day 1 Comprehensive Report | July 28, 2026

[3]:

```
# Dataset information
```

```
df.info()
```

```
<class 'pandas.DataFrame'>
```

```
RangeIndex: 10000 entries, 0 to 9999
```

```
Data columns (total 22 columns):
```

| #  | Column                    | Non-Null Count | Dtype   |
|----|---------------------------|----------------|---------|
| 0  | Property_ID               | 10000 non-null | str     |
| 1  | Property_Age_Years        | 10000 non-null | int64   |
| 2  | Property_Type             | 10000 non-null | str     |
| 3  | Location_Type             | 10000 non-null | str     |
| 4  | Property_Size_SqFt        | 10000 non-null | float64 |
| 5  | Property_Value_INR        | 10000 non-null | float64 |
| 6  | Construction_Type         | 10000 non-null | str     |
| 7  | Roof_Type                 | 9900 non-null  | str     |
| 8  | Number_of_Floors          | 10000 non-null | int64   |
| 9  | Occupancy_Type            | 10000 non-null | str     |
| 10 | Security_Level            | 8935 non-null  | str     |
| 11 | Natural_Disaster_Risk     | 10000 non-null | str     |
| 12 | Crime_Risk                | 10000 non-null | str     |
| 13 | Previous_Claims_Count     | 10000 non-null | int64   |
| 14 | Previous_Claim_Amount_INR | 9900 non-null  | float64 |
| 15 | Coverage_Amount_INR       | 10000 non-null | float64 |
| 16 | Deductible_INR            | 10000 non-null | float64 |
| 17 | Policy_Term_Years         | 10000 non-null | int64   |
| 18 | Customer_Age              | 10000 non-null | int64   |
| 19 | Customer_Income_INR       | 10000 non-null | float64 |
| 20 | Credit_Score              | 9900 non-null  | float64 |
| 21 | Total_Annual_Premium_INR  | 10000 non-null | float64 |

```
dtypes: float64(8), int64(5), str(9)
```

```
memory usage: 1.7 MB
```

[ ]:

Fig 2: DataFrame info displaying Non-Null counts and data types across all 22 columns.

## 3.2 Data Quality Inspection (Missing Values & Duplicates)

A critical phase of Day 1 was assessing the data quality. An analysis was performed to detect missing values across all columns. The findings indicate missing data in four specific features:

- Roof\_Type: 100 missing values
- Security\_Level: 1065 missing values (Highest proportion)
- Previous\_Claim\_Amount\_INR: 100 missing values
- Credit\_Score: 100 missing values

These missing values will require strategic imputation in Day 2. Furthermore, a duplication check revealed 0 duplicate rows, ensuring data integrity at the observation level.

# Machine Learning Project: Insurance Premium Prediction

Day 1 Comprehensive Report | July 28, 2026

[5]:

```
# Missing values  
print(df.isnull().sum())
```

|                           |      |
|---------------------------|------|
| Property_ID               | 0    |
| Property_Age_Years        | 0    |
| Property_Type             | 0    |
| Location_Type             | 0    |
| Property_Size_SqFt        | 0    |
| Property_Value_INR        | 0    |
| Construction_Type         | 0    |
| Roof_Type                 | 100  |
| Number_of_Floors          | 0    |
| Occupancy_Type            | 0    |
| Security_Level            | 1065 |
| Natural_Disaster_Risk     | 0    |
| Crime_Risk                | 0    |
| Previous_Claims_Count     | 0    |
| Previous_Claim_Amount_INR | 100  |
| Coverage_Amount_INR       | 0    |
| Deductible_INR            | 0    |
| Policy_Term_Years         | 0    |
| Customer_Age              | 0    |
| Customer_Income_INR       | 0    |
| Credit_Score              | 100  |
| Total_Annual_Premium_INR  | 0    |

dtype: int64

[ ]:

|

Fig 3: Output of `df.isnull().sum()` highlighting columns with missing data.

## 3.3 Feature Categorization & Statistical Summary

To prepare for feature engineering, the dataset's columns were segregated into numerical (int64, float64) and categorical (object/str) types. The descriptive statistics for numerical features revealed a wide variance in attributes like 'Property\_Value\_INR' and 'Customer\_Income\_INR', hinting at potential outliers that may need scaling.

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Day 1 Comprehensive Report | July 28, 2026

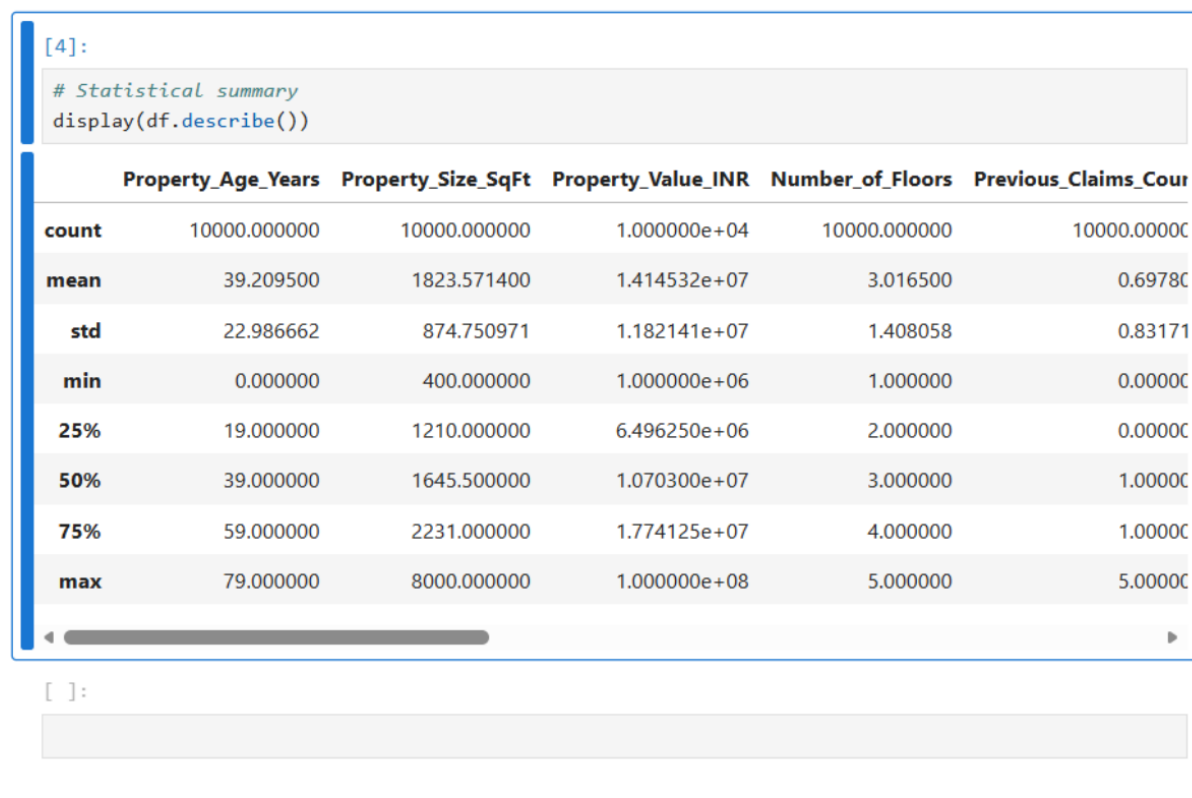


Fig 4: Statistical summary for numerical columns highlighting mean, std, and quartiles.

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Day 1 Comprehensive Report | July 28, 2026

```
[11]: # Numerical columns
print("Numerical columns:")
print(df.select_dtypes(include=["int64", "float64"]).columns.tolist())

Numerical columns:
['Property_Age_Years', 'Property_Size_SqFt', 'Property_Value_INR', 'Number_of_Floors', 'Previous_Claims_Count', 'Previous_Claim_Amount_INR', 'Coverage_Amount_INR', 'Deductible_INR', 'Policy_Term_Years', 'Customer_Age', 'Customer_Income_INR', 'Credit_Score', 'Total_Annual_Premium_INR']
```

Fig 5: Dynamic extraction of numerical column names.



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Day 1 Comprehensive Report | July 28, 2026

```
[14]: categorical_columns = df.select_dtypes(include=["str"]).columns.tolist()

print("Categorical columns:")
print(categorical_columns)

Categorical columns:
['Property_ID', 'Property_Type', 'Location_Type', 'Construction_Type', 'Roof_Type', 'Occupancy_Type', 'Security_Level', 'Natural_Disaster_Risk', 'Crime_Risk']
```

Fig 6: Categorical columns identified for future one-hot or label encoding.

## 3.4 Target Variable Analysis

The primary target for our regression model is 'Total\_Annual\_Premium\_INR'. A detailed univariate analysis was conducted on this specific variable.

Key Statistical Insights:

- Mean Premium: ~66,980 INR
- Median Premium (50%): 49,400 INR
- Minimum Premium: 5,000 INR
- Maximum Premium: 701,600 INR

The significant difference between the mean and median, along with the high maximum value, indicates a strong right-skewed distribution. This was visually confirmed via a histogram plot, where the majority of premiums are clustered below 100,000 INR, with a long tail extending towards higher values.

```
[15]: target = "Total_Annual_Premium_INR"
      display(df[target].describe())
```

|       |               |
|-------|---------------|
| count | 10000.000000  |
| mean  | 66980.980000  |
| std   | 60368.311833  |
| min   | 5000.000000   |
| 25%   | 28900.000000  |
| 50%   | 49400.000000  |
| 75%   | 83100.000000  |
| max   | 701600.000000 |

Name: Total\_Annual\_Premium\_INR, dtype: float64

Fig 7: Descriptive statistics isolated for the target variable.

```
name: Total_Annual_Premium_INR, dtype: float64
```

```
[16]: plt.figure(figsize=(10, 6))

plt.hist(df[target], bins=30)

plt.title("Distribution of Annual Insurance Premium")
plt.xlabel("Annual Premium (INR)")
plt.ylabel("Frequency")

plt.show()
```

Fig 8: Histogram of Total\_Annual\_Premium\_INR demonstrating a right-skewed distribution.

## 4. Strategic Roadmap for Day 2

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Building upon the foundational understanding established today, the following tasks are prioritized for Day 2:

- **Data Imputation:** Apply median imputation for skewed numerical features (e.g., `Credit_Score`) and mode imputation for categorical features (e.g., `Security_Level`).
- **Outlier Mitigation:** Investigate extreme values in the target variable and features using boxplots; apply log transformations if necessary to normalize the target distribution.
- **Feature Encoding:** Transform categorical variables into machine-readable formats using One-Hot Encoding.
- **Bivariate Analysis:** Perform correlation mapping to identify the features with the highest predictive power for the insurance premium.